



University of Moratuwa

B.Sc. Engineering

Biomechanics

Walking gait Analysis

Analysis of the walking gait through matlab

Laboratory: Biomechanics Laboratory

Assignment: Electromyography signal acquisition and interpretation

Student registration: 230449N

Sampling frequency: 100 Hz

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1 Introduction

Human gait is the pattern of movement of the limbs during locomotion. Gait analysis involves the systematic study of human motion through observational and quantitative measurement parameters. Understanding kinematics (motion trajectories, velocities, joint angles) and temporal-spatial parameters (gait cycle time, stance/swing ratio, stride length, cadence) is essential in clinical biomechanics, sports performance assessment, and rehabilitation engineering. This practical focuses on analyzing motion capture marker data collected from level walking to quantify gait characteristics.

2 Methods

Kinematic data were acquired using an optical motion capture system tracking passive reflective markers placed on lower limb anatomical landmarks (ASIS - ILIACA, PSIS - ILIACP, Greater Trochanter - HIP, Lateral Femoral Condyle - KNEE, Lateral Malleolus - ANKLE, and 2nd Metatarsal Head - TOE). Marker positions were recorded at a sampling rate of 100 Hz (0.01 s interval per frame). The recorded dataset `markers.csv` was processed using MATLAB script `gait230449N.m`.

Processing included:

1. Reconstructing 2D (x - z) and 3D (x - y - z) trajectories of pelvic and leg markers.
2. Tracking vertical (z) positions of the ankle and toe markers to identify key gait cycle events: Initial Heel Strike (HS_1), Foot Flat (FF), Heel Off (HO), Toe Off (TO), and Subsequent Heel Strike (HS_2).
3. Calculating temporal parameters (T_{cycle} , T_{stance} , T_{swing}) and spatial parameters (stride length, cadence, average progression velocity).
4. Evaluating instantaneous progression velocity from pelvic markers (x -coordinate derivative).
5. Comparing 2D and 3D thigh segment lengths to analyze planar approximation errors.
6. Computing 2D knee joint angle trajectories using the Law of Cosines on the hip-knee-ankle triangle.

2.1 MATLAB Implementation Code for Data Processing

The data extraction and preparation code implementation in MATLAB is given below:

```

1 % Load the data into array A (marker data)
2 A = csvread('markers.csv');
3
4 % Number of rows (frames) and columns in A
5 [rA,cA] = size(A);
6
7 % Data point (frame) numbers
8 datapt = A(:,1);
9
10 % Right side markers. Each is rA x 3 = [x y z] in mm.
```

```

11 iliaca = A(:, 5: 7);      % anterior superior iliac spine (ASIS)
12 iliaccp = A(:,11:13);    % posterior superior iliac spine (PSIS)
13 hip    = A(:,17:19);    % greater trochanter (TRO)
14 knee   = A(:,23:25);    % lateral femoral condyle (LFC)
15 ankle  = A(:,29:31);    % lateral malleolus (LMA)
16 toe    = A(:,35:37);    % metatarsal 2nd head (TOE)

```

Listing 1: Data initialization and setup in MATLAB (gait230449N.m)

```

1 %% Task 4 -- Gait cycle temporal calculations
2 HS1 = 204;  FF = 230;  HO = 258;  TO = 276;  HS2 = 314;
3
4 T_cycle = (HS2-HS1)/100;    % gait cycle time (s)
5 T_stance = (TO-HS1)/100;    % stance time (s)
6 T_swing = (HS2-TO)/100;    % swing time (s)
7 P_stance = (T_stance / T_cycle) * 100;
8 P_swing = (T_swing / T_cycle) * 100;
9
10 %% Task 5 -- Stride length, cadence, average velocity
11 idx_HS1 = HS1 - datapt(1) + 1;
12 idx_HS2 = HS2 - datapt(1) + 1;
13 stride_length = (ankle(idx_HS2,1) - ankle(idx_HS1,1))/1000; % m
14 cadence = (2 * 60) / T_cycle; % steps/min
15 v_avg = stride_length / T_cycle; % m/s
16
17 %% Task 6 -- Instantaneous velocity
18 inst_vel_iliaccp = ((iliaccp(2:rA,1)-iliaccp(1:rA-1,1))/0.01)/1000;
19 inst_vel_iliaca = ((iliaca(2:rA,1)-iliaca(1:rA-1,1))/0.01)/1000;
20
21 %% Task 7 -- Thigh segment length comparison
22 thigh_len_3D = sqrt((hip(:,1)-knee(:,1)).^2 + (hip(:,2)-knee(:,2)).^2 +
23 (hip(:,3)-knee(:,3)).^2);
24 thigh_len_2D = sqrt((hip(:,1)-knee(:,1)).^2 + (hip(:,3)-knee(:,3)).^2);
25
26 %% Task 8 -- 2D Knee Joint Angle
27 thigh_len = thigh_len_2D;
28 leg_len = sqrt((knee(:,1)-ankle(:,1)).^2 + (knee(:,3)-ankle(:,3)).^2);
29 hip_ankle_len = sqrt((hip(:,1)-ankle(:,1)).^2 + (hip(:,3)-ankle(:,3)).^2);
30 knee_angle = 180 - acosd((thigh_len.^2 + leg_len.^2 - hip_ankle_len.^2) ./ (2 .* thigh_len .* leg_len));

```

Listing 2: MATLAB code for kinematic calculations, event detection, velocity and knee angle estimation

3 Results

3.1 Gait Cycle Events and Temporal Parameters

From the vertical displacement trajectories of ankle and toe markers (Figure 3), key gait events were identified at the following data point frames:

- **Initial Heel Strike (HS₁):** Frame 204 ($t = 2.04$ s)
- **Foot Flat (FF):** Frame 230 ($t = 2.30$ s)

- **Heel Off (HO):** Frame 258 ($t = 2.58$ s)
- **Toe Off (TO):** Frame 276 ($t = 2.76$ s)
- **Subsequent Heel Strike (HS₂):** Frame 314 ($t = 3.14$ s)

Table 1: Calculated Gait Parameters for Subject 230449N

Parameter	Value	Unit
Gait Cycle Duration (T_{cycle})	1.10	s
Stance Duration (T_{stance})	0.72	s
Swing Duration (T_{swing})	0.38	s
Stance Percentage (P_{stance})	65.45	%
Swing Percentage (P_{swing})	34.55	%
Stride Length	1.353	m
Cadence	109.09	steps/min
Average Velocity (v_{avg})	1.230	m/s

3.2 MATLAB Output Figures and Analysis

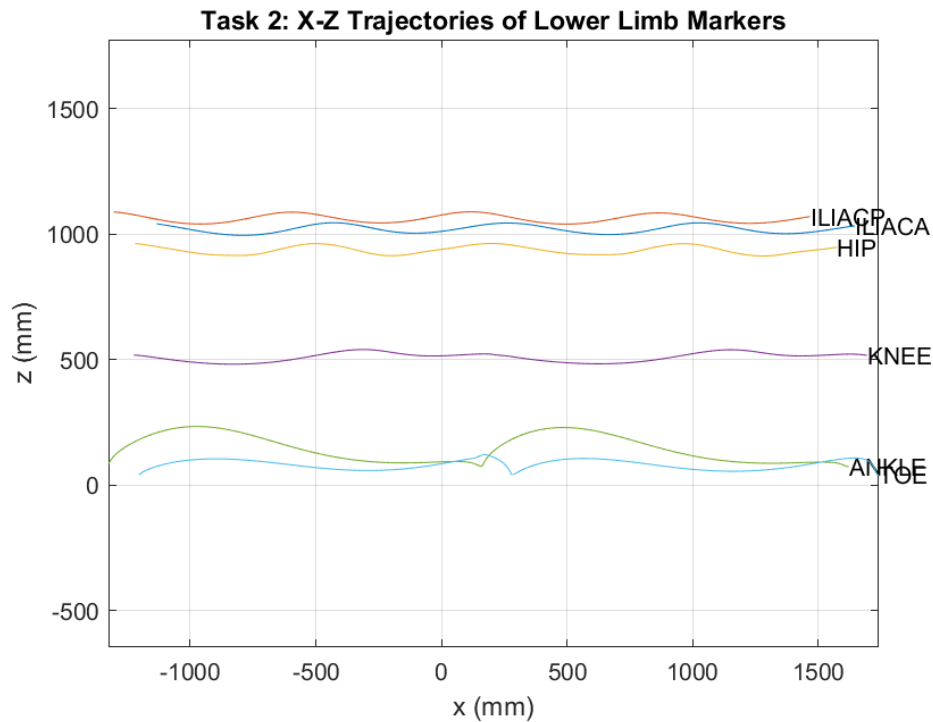


Figure 1: Task 2: x - z sagittal plane trajectories of pelvic, hip, knee, ankle, and toe markers during progression.

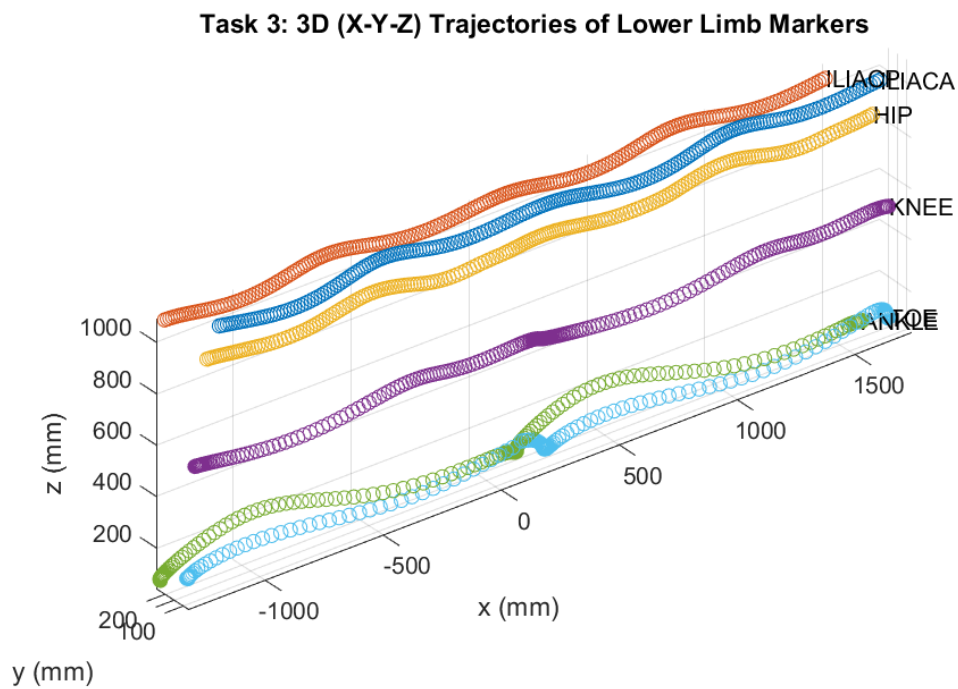


Figure 2: Task 3: 3D (x - y - z) spatial trajectories showing out-of-plane movements of lower limb markers.

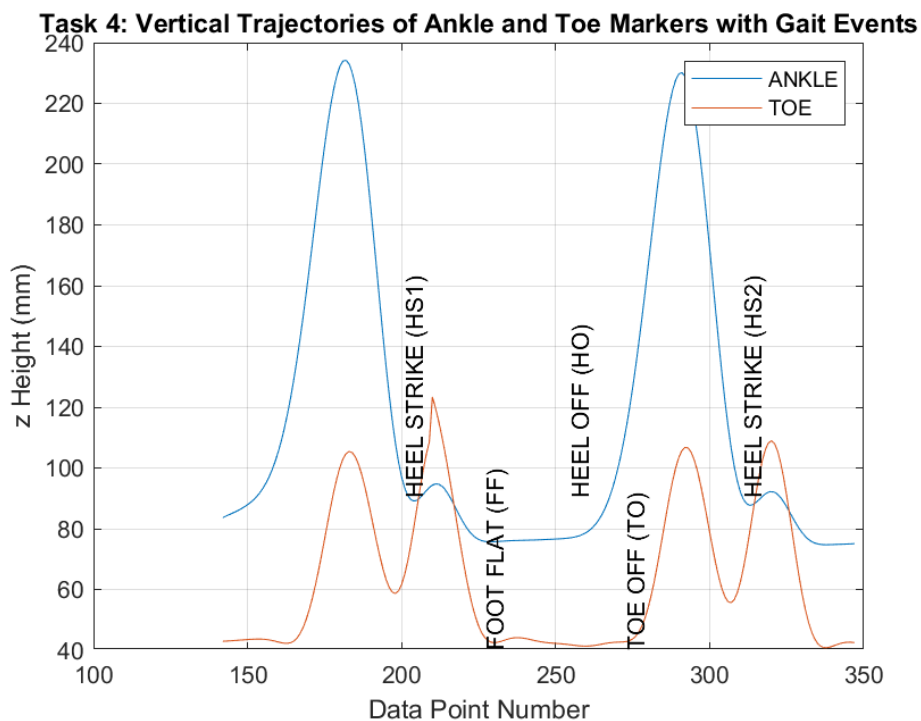


Figure 3: Task 4: Vertical (z) height trajectories of ankle and toe markers with annotated key gait cycle events (HS_1 , FF, HO, TO, HS_2).

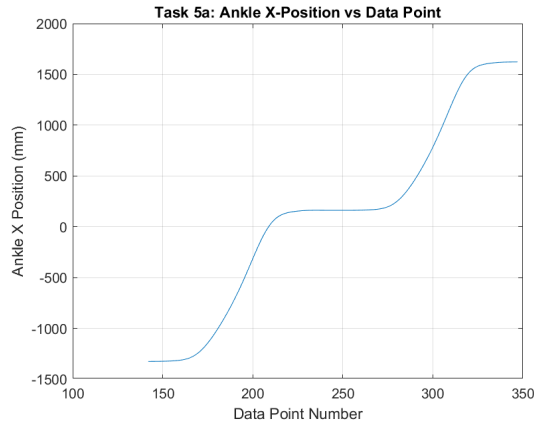
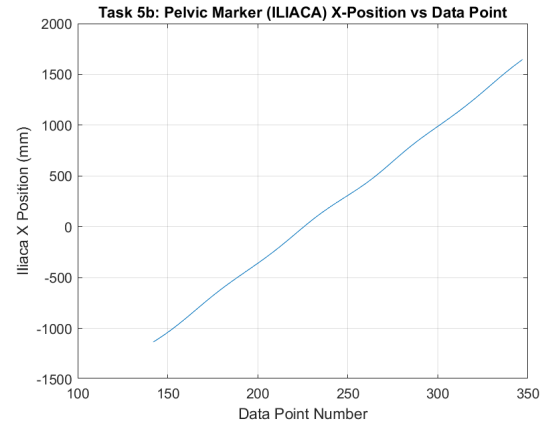
(a) Ankle x -position vs frame(b) Pelvis (ILIACA) x -position vs frame

Figure 4: Task 5: Forward displacements (x -coordinates) of ankle and pelvic markers used to determine stride length.

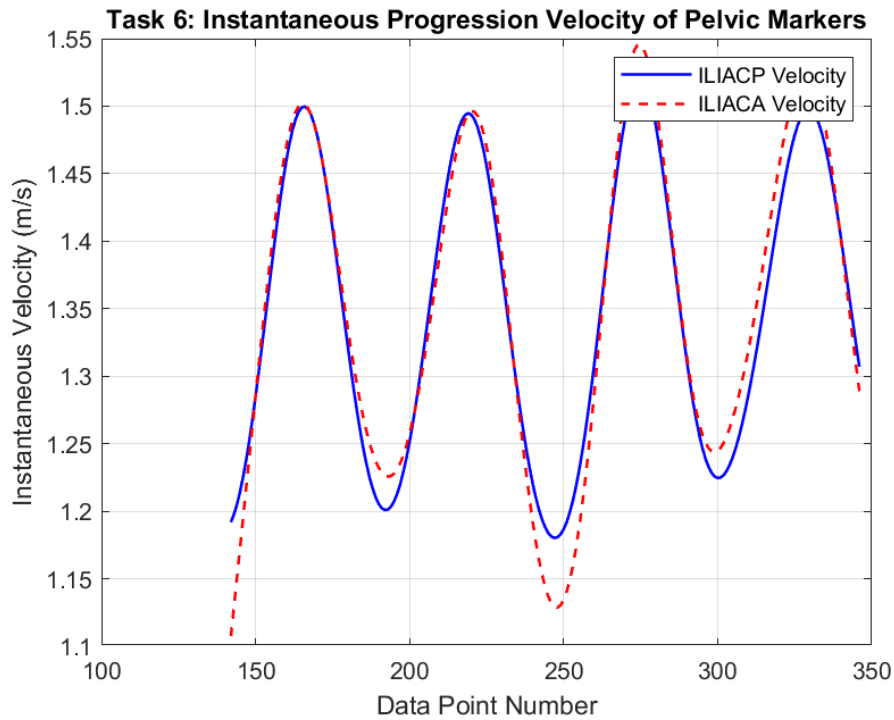


Figure 5: Task 6: Instantaneous progression velocity of pelvic markers (ILIACP and ILIACA) derived frame-by-frame.

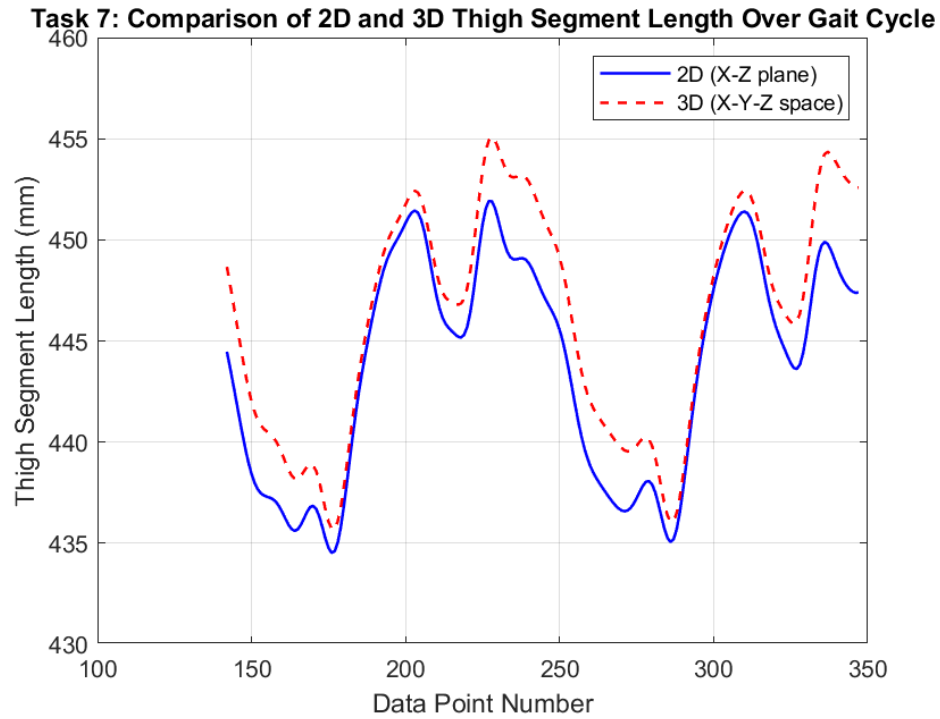


Figure 6: Task 7: Comparison of thigh segment length calculated in 2D (x - z) vs 3D (x - y - z) space.

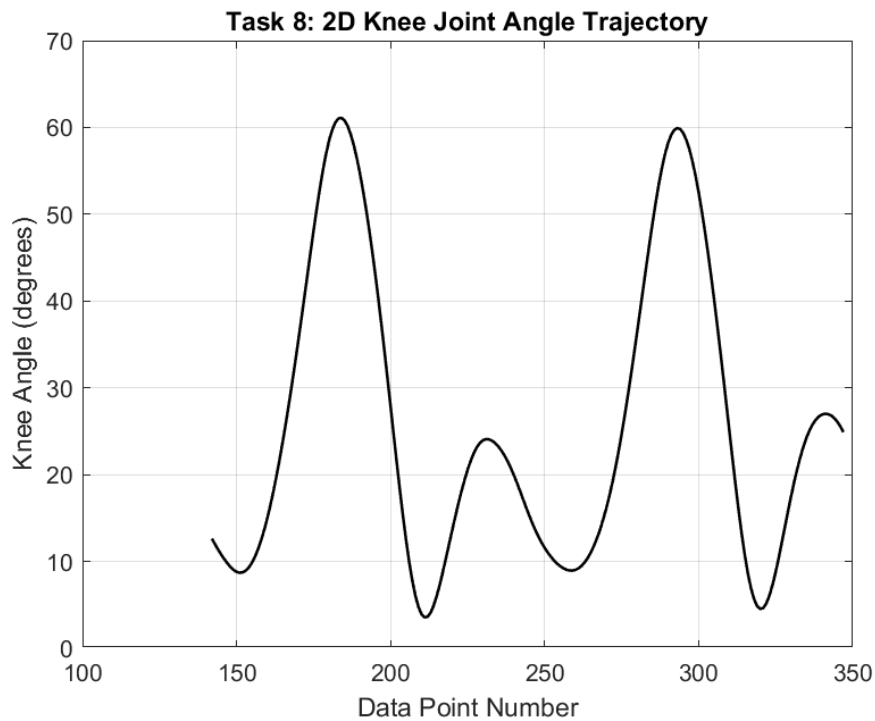


Figure 7: Task 8: 2D Knee joint angle trajectory over the gait cycle showing stance and swing flexion peaks.

4 Discussion of Results

4.1 Identification of Heel Strike and Toe Off

Heel strike occurs when the heel makes initial contact with the ground, corresponding to local minima in ankle height combined with zero or low vertical velocity. Foot flat follows as the forefoot contacts the ground. Heel off is observed when the ankle height begins to rise sharply while toe height remains low. Toe off occurs when the toe marker loses contact with the ground and accelerates upwards into the swing phase.

4.2 Measurement Sources of Error

1. **Soft Tissue Artifact (STA):** Skin and muscle movement over underlying bones causes relative motion between optical markers and skeleton landmarks.
2. **Anatomical Landmark Placement Error:** Misalignment of markers relative to exact joint rotation centers (e.g., greater trochanter or condyle center).
3. **2D Planar Projection Error:** Estimating parameters purely in 2D (x - z plane) ignores pelvic rotation and hip abduction/adduction (y -axis motion), leading to small deviations in segment length and knee angle calculations.

5 Conclusion

In this laboratory session, quantitative gait analysis was conducted using optical marker trajectories processed in MATLAB. Key gait cycle phases, temporal-spatial parameters, instantaneous velocities, and knee angle kinematics were successfully derived. The results demonstrate classic level-walking gait mechanics ($\sim 65\%$ stance and $\sim 35\%$ swing duration) and highlight the importance of 3D kinematic modeling to account for out-of-plane joint motion.